

Software Requirements Specification (SRS)

Autonomous Multi-Agent B2B Negotiation Network

Document Version: 2.0

Primary Runtime: Python 3.10+ (FastAPI)

AI Platform: Alibaba Cloud Model Studio (Qwen 3.x)

Messaging Channel: WhatsApp Business Platform Cloud API

Status: Engineering Blueprint / Production Roadmap

Frontend: HTML5 + CSS3 + JS (ES6+)

Marketplace Platform: Alibaba.com Open Platform

Cloud Ecosystem: Alibaba Cloud Infrastructure

1. Executive Summary

The **Autonomous Multi-Agent B2B Negotiation Network** is an enterprise-grade AI-powered procurement and sales automation platform. It enables businesses to configure custom commercial negotiation boundaries and delegate complete B2B negotiations to autonomous AI agents. Rather than relying on tedious manual communication via email, calls, or marketplace chat, the system instantiates isolated *Buyer* and *Seller* AI agents operating under strict commercial guardrails.

Agents negotiate autonomously according to predefined commercial rules and parameters including commodity type, quantity, unit, starting price, reservation price, delivery deadline, payment terms, quantity flexibility, shipping requirements, strategy persona, maximum rounds, and walkaway conditions.

Architecture Evolution (v2.0)

This revised version replaces legacy Streamlit prototypes with an enterprise stack: custom HTML5/CSS/JS frontend, FastAPI backend, official WhatsApp Business Platform Cloud API as the communication transport, and Alibaba Cloud (Model Studio / Qwen) as the primary cloud and AI ecosystem.

2. Project Vision & Workflow

The long-term vision is to create a seamless autonomous B2B trading network that handles the entire transaction lifecycle:

Human Business → Web Dashboard → Negotiation Orchestrator → Buyer AI / Seller AI →
WhatsApp Cloud API → Counterparty

The target end-to-end operational pipeline encompasses:

Product Discovery → Supplier Discovery → Supplier Verification → Price/Quantity/Delivery
Negotiation → Final Offer → Approval → Contract → Order → Tracking → Audit

3. Revised Technology Stack

3.1 Frontend Architecture

The frontend is built cleanly using standard web technologies: HTML5, CSS3, JavaScript (ES6+), Fetch API, and Chart.js for live visual updates via Server-Sent Events / WebSockets.

Main Application Interfaces:

- /login & /dashboard: Authentication and operational overview
- /negotiations & /negotiations/:id: Active negotiation monitor and detailed history
- /create-negotiation: Parametric constraint configuration form
- /suppliers, /products, /contracts, /orders, /audit, /settings

3.2 Alibaba Cloud Infrastructure Blueprint

Requirement Layer	Alibaba Cloud / Core Technology Component
AI / LLM Runtime	Alibaba Cloud Model Studio / Qwen 3.x (OpenAI-compatible API)
Backend API Gateway & Compute	Alibaba Cloud API Gateway + FastAPI on ECS / ACK Cluster
Database (Transactional)	ApsaraDB RDS PostgreSQL
Caching & State Management	Tair (Redis-compatible High Performance Engine)
Object & Document Storage	Object Storage Service (OSS)
Immutable Audit Trail	OSS WORM (Write Once Read Many) Retention Policies
Secrets & Cryptography	Key Management Service (KMS) & Secrets Manager
Messaging & Events	EventBridge & Function Compute (Serverless)
Observability & Logging	Simple Log Service (SLS)
External Channels	Alibaba.com Open Platform & WhatsApp Business Cloud API
Orchestration & Validation	LangGraph & Pydantic v2 Validation Pipelines

4. AI Layer & Model Integration

The AI decision engine leverages **Alibaba Cloud Model Studio**, utilising high-capability Qwen models (e.g., qwen3.7-max and qwen3.7-plus). The system adopts a dual-model strategy:

- **Fast/Cost-effective Model:** Handles routine round calculations and basic message formatting.
- **High-Capability Model:** Handles complex multi-variable trade-offs, initial counterparty strategy analysis, and contract risk evaluation.

5. Alibaba.com & WhatsApp Integration

5.1 Alibaba.com Open Platform Integration

Alibaba.com serves as the primary marketplace data layer through its Open Platform developer model (App Key / App Secret + OAuth authorization). Standard capabilities include:

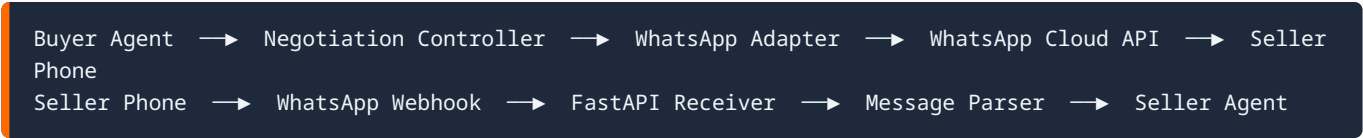
- **Supplier & Product Discovery:** Querying supplier profiles, verification credentials, and catalog data.
- **Supplier Scoring Function:**

Supplier Score = Price Score + Quality Score + Supplier Score + Delivery Score + Historical Performance

- **Data Normalization:** Mapping marketplace responses directly into internal Pydantic product schemas.

5.2 WhatsApp Communication Channel

WhatsApp acts as the external communications transport while the core negotiation logic remains fully isolated within the secure backend.



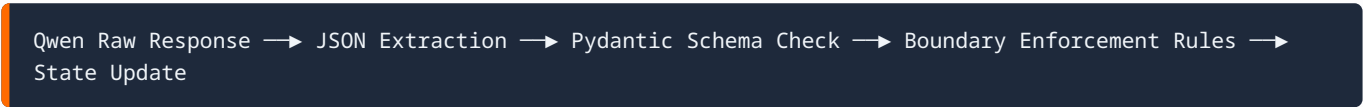
6. Security, Agent Isolation & State Management

Critical Security Requirement: Agent Privacy Isolation

The server enforces strict physical and logical state isolation between negotiating counterparties. Private reservation prices, private strategy memory, walkaway thresholds, and internal scoring metrics are **NEVER** exposed to the opposing agent or sent over public communications.

6.1 Strict Boundary Validation Pipeline

No LLM output is allowed to alter financial or business state directly. All model responses pass through a strict validation pipeline:



7. Functional Requirements (FR)

- **FR-01 (Business Registration):** Register company profile, contact details, currency, and WhatsApp credentials.
- **FR-02 (User Authentication):** Enforce secure RBAC, password hashing, and multi-tenant isolation.
- **FR-03 (Negotiation Setup):** Define starting price, reservation boundary, rounds, delivery, payment terms.
- **FR-04/05 (Supplier Discovery & Ranking):** Programmatically search and rank Alibaba.com suppliers using normalized scoring criteria.
- **FR-06/07 (Agent Instantiation & Offer Generation):** Instantiate isolated agents generating structured JSON counter-offers.
- **FR-08/09 (JSON & Boundary Enforcement):** Validate structured JSON and strictly block out-of-bound commercial offers.
- **FR-10 (Convergence Settlement):** Midpoint Settlement:
$$Final\ Price = (Buyer\ Offer + Seller\ Offer) / 2 \quad (\text{when } Buyer\ Offer \geq Seller\ Offer)$$
- **FR-16/17 (Contract Generation & Immutable Audit):** Generate PDF contracts, hash contract data, sign via KMS, and store in Alibaba Cloud OSS with WORM retention rules.

8. Project Implementation Roadmap

Phase	Duration	Key Deliverables
Phase 0: Cloud Setup	1 Week	Alibaba Cloud account provisioning, RAM roles, RDS/Tair/OSS/KMS setup, WhatsApp Business verification.
Phase 1: Frontend Development	1-2 Weeks	HTML5/CSS3/JS Web Dashboard, Chart.js integrations, negotiation monitoring views.
Phase 2: Backend Core	1-2 Weeks	FastAPI scaffolding, PostgreSQL schema migrations, JWT auth, CRUD endpoints.
Phase 3: AI Engine Integration	1 Week	Model Studio Qwen client, structured output validation, latency/prompt benchmarking.
Phase 4: Negotiation Controller	2 Weeks	LangGraph orchestration state machine, mathematical concession rules, boundary guardrails.
Phase 5: WhatsApp Adapter	1-2 Weeks	Webhook handlers, Cloud API outbound delivery, event parsing, message deduplication.
Phase 6: Alibaba Marketplace	2-4 Weeks	Open Platform OAuth, supplier discovery, catalog retrieval, automated ordering APIs.
Phase 7: Audit & Legal Contracts	1-2 Weeks	PDF contract generator, KMS digital signing, OSS WORM retention enforcement.
Phase 8: Production Deployment	1-2 Weeks	ACK/ECS deployment, WAF configuration, SLS logging pipelines, end-to-end security audit.